

Experiment 3 – Implement a Bayesian Network to Diagnose COVID-19 Infection Using a Standard Dataset

Problem Statement

A public health analytics team wants to develop a machine learning model to predict the COVID-19 test result of an individual based on demographic information, symptoms, and test-related information.

Collect **Israel COVID-19 Dataset (2020–2021)** available at, <https://www.kaggle.com/datasets/mykeysid10/covid19-dataset-for-year-2020/data> and **Construct a Bayesian Network and perform probabilistic inference to diagnose COVID-19 infection based on the observed evidence.**

Task 1: Dataset Investigation

Load the Israel COVID-19 dataset and investigate:

1. Number of records and attributes
2. Missing values
3. Target variable
4. Distribution of COVID-19 results

Task 2: Data Preprocessing

- Select relevant COVID-19 symptoms
- Handle missing values
- Convert variables into appropriate discrete categories.
- Prepare the dataset for Bayesian Network construction

Task 3: Construct a Bayesian Network using at least four relevant variables from the dataset.

The network must include:

- At least one variable representing patient characteristics.
- At least two symptom-related variables.
- The COVID-19 test result/infection status as the outcome variable.

Justify the relationships between the selected variables and the outcome variable.

Task 4: Learn the Probability Distributions

Using the preprocessed dataset:

1. Estimate the prior probabilities of the relevant variables.
2. Learn the conditional probability distributions required by the Bayesian Network.
3. Display the conditional probability table associated with the COVID-19 outcome.

Task 5: Perform Probabilistic Inference

Given the following patient evidence:

Fever = Yes

Cough = Yes

Sore Throat = No

Age 60+ = Yes

Use the Bayesian Network to infer:

$$P(\text{COVID Positive} \mid \text{Evidence})$$

and

$$P(\text{COVID Negative} \mid \text{Evidence})$$

Based on the posterior probabilities, determine the most probable diagnosis.

Task 6: What-If Analysis

Change one piece of evidence:

Fever = Yes $\rightarrow$ Fever = No

Compare the probability:

$$P(\text{COVID Positive} \mid \text{Evidence})$$

before and after changing the fever status.

How did changing the evidence affect the inferred probability of COVID-19 infection?

Explain the result based on the Bayesian Network.